Disease briefing | Risk assessment | D2R2 | Decision support | Ranking

Animal & Plant Health Agency

Data Quality Statement for

Disease briefing, Decision support, Ranking and Risk Assessment tool (D2R2)

Introduction: This statement provides an overview of the quality of the data within the D2R2 database based on the seven primary dimensions for data quality assessment. This statement is written in the context of the data within D2R2 being used to support disease briefing, decision support, disease ranking and risk assessment provided by APHA. The database is stored and maintained on an APHA server to which only a restricted number of APHA users have access. All the data are collated by APHA staff.

Database Title: **D2R2** (**D**isease briefing, **D**ecision support, **R**anking and **R**isk assessment)

Overview and purpose of database: D2R2 contains a wide range of standardised information about exotic and endemic diseases held in 'profiles'. Each profile includes 40 scored criteria agreed with policy makers. D2R2 provides relative rankings for every disease profiled, a range of reports for each disease and the functionality to explore the impact of changes in any criterion on the ranking and score for a disease (What if? scenarios). These outputs aid the prioritisation and management of animal issues and diseases by government, helping to guide resource allocation towards diseases likely to have a greater impact on one of the four reasons for government intervention (RFIs) described in the GB Animal Health and Welfare Strategy (i.e. animal welfare, public health, international trade, and wider society) and/or a higher probability of occurrence. D2R2 uses simplified multi-criteria decision analysis methodology and relies on evidence compiled by experts and from published material wherever possible. It allows for efficient use of expertise, as standardised, maintained briefing material, reducing the need for on-call, reactive expert input for policy development, and enables rapid access to information e.g. during outbreaks of exotic disease.

Summary of information it holds: The database contains standardised information on exotic and endemic diseases. Each profile consists of sixteen sections including epidemiology, surveillance, risk assessment, the four reasons for government intervention (RFIs, which are public health, welfare, wider society and international trade), legislation and the approach to control.

Population of D2R2 profiles: Disease profiles are completed by a disease expert, usually from APHA or the Pirbright Institute using published evidence where possible. Where expert opinion is used this is made explicit. Each question has guidance for authors to explicitly define the criterion and each categorical option to minimise inconsistencies. Authors have restricted access to the D2R2 database, limited to the profiles that they are responsible for.

Validation of D2R2 profiles: Colleagues independent of the profile's author check responses to scoring questions and accuracy of evidence within the profile. Scientific editorial review is conducted to check each question answered is in line with guidance. The profile is then reviewed by a government veterinarian and the policy lead. Once the tripartite validation process has been completed, the profile is then published on the APHA intranet.

Extraction of Data from the database: Intranet access available to all Defra/APHA colleagues enables users to view full published profile content and printable pdf reports providing disease information, briefing material and ranking reports.

Use of D2R2 data: The information contained within D2R2 will assist with the prioritisation and management of animal issues and diseases by Government, helping to guide resource allocation towards diseases likely to have a greater impact and/or a higher probability of occurrence. It also provides standardised disease information to assist with drafting briefing or to provide background information to staff working on a particular disease or issue.

Data retention: Data about diseases and guidance on how to use D2R2 will be regularly reviewed (maximum review interval of 5 years for areas least likely to need change) but provides an ongoing source of information and disease ranking. As disease profiles and ranking reports are updated previous versions are automatically stored as an archive on the system for the lifetime of the database.

Category (+definition)	Quality description
Relevance of data	- **what information is used**: The author must quote the source of the evidence provided, where possible using published material. The database captures all references including legislation and further information sources. Where expert opinion is used this must be made explicit. When diseases for which there is little information are first profiled (e.g. new diseases such as Schmallenberg virus when first identified), the expert is asked to use their judgment, and to select the most likely option. In all cases if the expert is unsure between two categories, they are asked to be precautionary, and select the 'worst', option. All evidence reflects the current situation, i.e. the level and effects of the disease as it currently occurs in GB (or estimated most likely situation after an exotic disease incursion), given current control measures. Where there are differences between English, Welsh and Scottish Government policies this is recorded, and the option that reflects English policy is selected.
Timeliness [the degree to which data represent reality from the required time point]	-**how often are disease profiles commissioned:** Currently, two new disease profiles are commissioned each year. The decision on which profiles to commission depends on policy appetite. -**when do the profiles become available**: Once the profile has been written by the disease expert, the profile then currently undergoes a tripartite review system, which includes scientific, veterinary and policy

	review. The duration of time taken to publish a profile depends on the availability of resource to validate and review each profile. -**how often are the profiles reviewed:** Usually every two years, but some sections will have a 5 year interval, whilst others may be reviewed more frequently for certain diseases. If policy has changed or there's been a recent outbreak or a new diagnostic test, legislation or control method developed this information should be promptly added to the disease profile rather than awaiting the next scheduled review. -**how often are data contained in the profiles updated**: UK trade data for the previous year are inputted into the D2R2 database on an annual basis. Zoonotic disease incidence/prevalence data from UK Zoonoses report from the previous year are inputted into the relevant disease profiles on an annual basis. Data from the international disease monitoring team on the risk of incursion of exotic disease are inputted into the relevant profiles on a quarterly basis or more frequently if the situation changes
Accuracy and precision [extent of data error and bias and how well data portrays reality]	-**how are the profiles completed**: Profiles are written and completed by recognised disease experts following D2R2 guidance documents. The profile reflects the current state of knowledge about the disease being profiled based on the scientific literature and/or expert opinion. -**how are the ranking scores generated:** D2R2 calculates a score for each criterion, derived from the score for the categorical option chosen, multiplied by the weighting for that criterion. The values for each criterion within each section are summed to give a summary score that describes that disease's impact on the four reasons for government intervention (RFIs) (i.e. Animal welfare, public health, international trade, wider society) and also for the 'Post mitigation risk'. D2R2 transforms each of these five scores generated for each disease into a percentage derived from a scale generated by allocating '0' for the value generated by summing the lowest value possible for each criterion, and '100' for that derived by summing the highest possible values. D2R2 reports scores for each section separately/independently and makes no assumption about the relative importance of the four reasons for government intervention. However as the highest ranked diseases for different RFIs may have numerically different scores, outputs are presented as ranked positions, to avoid implying that one RFI is more important than another. -**bias:** To minimise inadvertent bias the tool does not reveal the scores or weightings to the author.
Comparability [how well these data can be compared with data taken from the same dataset and with similar data from other sources]	-**within dataset comparability:** The database holds every version of each disease profile so that comparisons can be made between new and archived versions. **-other dataset comparability:** There are no other datasets within APHA holding similar data.
Coherence	-**disparate sources:** UK trade data are derived from the APHA trade

[degree to which data can be or have been merged with other data sources]	team at Carlisle (and include data derived from the EU Commission Trade Control Experts System (TRACES) records, from the Trade Statistics, Economics and Statistics Programme at York (and include trade statistics collected by HM Revenue and Customs). Disease incidence/prevalence data for Great Britain are derived from the UK Zoonoses Report for all zoonotic diseases and from other surveillance for the remainder (primarily from the surveillance activities of the Animal and Plant Health Agency (APHA) and the Scottish Agricultural College Consulting, Veterinary Services (SACCVS, operating within Scotland's Rural College – SRUC), using data gathered by the network of diagnostic laboratories. Quantitative diagnostic data for all of GB is provided by the Veterinary Investigation Diagnostic Analysis (VIDA) surveillance system. Information on the risk of disease incursion is provided directly from the APHA International Disease Monitoring (IDM) team. -**differences in datasets:** Trade data, incidence/prevalence data from the UK Zoonoses Report and from VIDA and data regarding the risk of incursion are taken directly from the original sources and so should be identical.
Interpretability [how well the data is understood and utilised appropriately]	-**Is there a particular context that this data needs to be considered within:** The data are collected for the specific purpose of creating automatically generated pdf documents that can be printed from D2R2, which currently include: i. A summary prioritisation report for each disease (provides each RFI score and its justification, a description of the disease, and the score for each of the contributing criteria, information about public perception, and uncertainty around the evidence, government costs and relevant legislation). ii. Summary profile iii. Full profile iv. Question and answer (Q&A) brief v. A disease ranking report **-what other information is available to help users better understand this data source:** There is guidance for authors on how to answer each question. This is available to all users within D2R2 in the 'Need help?' folder. Also in this folder are some more general guidance documents covering the approach to take to drafting a disease profile and to help users know how to navigate and use the available D2R2 functionality. The disease ranking report includes some explanatory text to help users interpret the information provided. PDF reports providing all the guidance available to authors on how to answer each question that is included in each PDF report on a particular disease (reports i, ii, iii and iv above) is also available for readers and other users of these outputs. This information is also directly accessible via an icon on each question to anyone browsing the profile electronically.
Accessibility	-**what data are shared and with whom**: All intranet published D2R2 disease profiles, disease ranking reports and epidemiological profiles are

[availability of relevant information and access to the data in a convenient and suitable manner]	available to all Defra/APHA colleagues via the D2R2 intranet page: http://vla65/SurveillanceProfiles/Search.aspx -**contact details for database queries**: all queries can be directed to D2R2@defra.gsi.gvo.uk